

Natasha S. Abrams
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EDUCATION

University of California, Berkeley, PhD
Astrophysics

Berkeley, CA
Aug 2021 – Expected May 2027

University of California, Berkeley, MA
Astrophysics

Berkeley, CA
Aug 2021 – May 2023

Harvard University, AB
Astrophysics and Physics
Magna Cum Laude with Highest Honors in Astrophysics and Physics

Cambridge, MA
Aug 2017 – May 2021

RESEARCH EXPERIENCE

Berkeley Fellow
University of California, Berkeley
Graduate Student Researcher
TVS Microlensing Subgroup; Vera C. Rubin Observatory

Aug 2021 – Present
Advisor: Jessica Lu
Oct 2020 – Present

Herchel Smith Fellow
Harvard University

Sept 2019 – June 2021
Advisor: Christopher Stubbs

Reischauer Summer Science Undergraduate Research Fellow
Kavli Institute for the Physics and Mathematics of the Universe – University of Tokyo
Undergraduate Researcher

June 2019 – Nov 2020
Advisor: Masahiro Takada
June 2017 – Aug 2017

Harvard University

Advisor: Allyson Bieryla

Undergraduate Researcher
Center for Astrophysics | Harvard & Smithsonian

Sept 2018 – May 2019
Advisors: Belinda Wilkes and Mojegan Azadi

PRISE Fellow
Center for Astrophysics | Harvard & Smithsonian
Researcher

June 2018 – Aug 2018
Advisor: Akos Bogdan
June 2015 – Jan 2016

American Museum of Natural History

Advisor: Arieh Maller

SOFTWARE DEVELOPMENT

PopSyCLE - *microlensing population synthesis code*; major contributor and primary maintainer
PUZLE - *ZTF microlensing pipeline*; major contributor and primary maintainer
BAGLE - *microlensing modeling software*; major contributor
SPISEA - *simple stellar population code*; major contributor
KAI - *Keck adaptive optics imaging reduction pipeline*; major contributor - lead pipeline modernization
flystar - *Precision relative astrometry code*; minor contributor
rubin_sim - *Rubin scheduler and cadence optimizer*; minor contributor
LightCurveLynx - *Lightcurve simulation tool*; minor contributor

PUBLICATIONS

3 first author, 1 co-first author, 16 coauthor, 1 as primary research mentor

* Directly supervised student

First Author

4. **Abrams, N. S.** et al. “Assessing the Impact of Binary Systems on Microlensing Using SPISEA and PopSyCLE Population Simulations”. *ApJ*. 980. P. 103. 2025
3. **Abrams, N. S.** et al. “Microlensing Discovery and Characterization Efficiency in the Vera C. Rubin Legacy Survey of Space and Time”. *ApJS*. 276. P. 10. 2025

2. Medford, M. S. & **Abrams, N. S.**, et al., “60 Microlensing Events from the Three Years of Zwicky Transient Facility Phase One.” *ApJ* 947 1, 2023.
1. **Abrams, N. S.**, and Takada, M. “Hunting Gravitational Wave Black Holes with Microlensing.” *The Astrophysical Journal*, vol. 905, no. 2, Dec. 2020, p. 121.

Coauthor

17. Huston, Macy J. et al. (*incl. Abrams, N. S.*). “An Updated SYNTHPOP Model for Microlensing Simulations. I. Model Description and Evaluation”. *AJ*. 172. P. 94. 2026
16. Wagg, Tom et al. (*incl. Abrams, N. S.*). “Charting the Galactic Underworld I: Comprehensive simulations of the kinematics, rates, and demographics of Milky Way black holes”. *arXiv e-prints*. arXiv:2607.22814. 2026
15. Begbie, Caitlin et al. (*incl. Abrams, N. S.*). “Expanding the SPISEA Stellar Population Synthesis Software to the Substellar Regime”. *arXiv e-prints*. arXiv:2607.14292. 2026
14. Lu, J. R. et al. (*incl. Abrams, N. S.*). “The BAGLE Python Package for Bayesian Analysis of Gravitational Lensing Events”. *AJ*. 172. P. 30. 2026
13. * Bhadra, T. Dex, Lu, J. R., **Abrams, N. S.**, et al. “Modeling Binary Lenses and Sources with the BAGLE Python Package”. *ApJ*. 1002. P. 60. 2026
12. Perkins, Scott E. et al. (*incl. Abrams, N. S.*). “Microlensing Constraints on Primordial Black Holes toward the Bulge Are Galactic Model Limited”. *ApJ*. 1006. P. 110. 2026
11. Terry, Sean K. et al. (*incl. Abrams, N. S.*). “Predictions of the Nancy Grace Roman Space Telescope Galactic Exoplanet Survey. IV. Lens Mass and Distance Measurements”. *AJ*. 171. P. 212. 2026
10. Miller, Adam A. et al. (*incl. Abrams, N. S.*). “The La Silla Schmidt Southern Survey”. *PASP*. 137. P. 094204. 2025
9. Kaczmarek, Zofia et al. (*incl. Abrams, N. S.*). “On Finding Black Holes in Photometric Microlensing Surveys”. *ApJ*. 981. P. 183. 2025
8. Fardeen, James et al. (*incl. Abrams, N. S.*). “Astrometric Microlensing by Primordial Black Holes with the Roman Space Telescope”. *ApJ*. 965. P. 138. 2024
7. Rowan, Dominick M. et al. (*incl. Abrams, N. S.*). “A hidden population of massive white dwarfs: two spotted K + WD binaries”. *MNRAS*. 529. Pp. 587–603. 2024
6. Perkins, Scott E. et al. (*incl. Abrams, N. S.*). “Disentangling the Black Hole Mass Spectrum with Photometric Microlensing Surveys”. *ApJ*. 961. P. 179. 2024
5. Street, R. A. et al. (*incl. Abrams, N. S.*). “LSST Survey Strategy in the Galactic Plane and Magellanic Clouds”. *ApJS*. 267. P. 15. 2023
4. Azadi, Mojegan et al. (*incl. Abrams, N. S.*). “Disentangling the AGN and Star formation Contributions to the Radio-X-Ray Emission of Radio-loud Quasars at $1 < Z < 2$ ”. *ApJ*. 945. P. 145. 2023
3. Rose, Sam et al. (*incl. Abrams, N. S.*). “The Impact of Initial-Final Mass Relations on Black Hole Microlensing”. *ApJ*. 941. P. 116. 2022
2. Lam, Casey Y. et al. (*incl. Abrams, N. S.*). “An Isolated Mass-gap Black Hole or Neutron Star Detected with Astrometric Microlensing”. 933. P. L23. 2022
1. Lam, Casey Y. et al. (*incl. Abrams, N. S.*). “Supplement: “An Isolated Mass-gap Black Hole or Neutron Star Detected with Astrometric Microlensing” (2022, *ApJL*, 933, L23)”. *ApJS*. 260. P. 55. 2022

WHITE PAPERS/TECHNICAL NOTES

4 first author, 8 coauthor
 * First or co-first author

12. Kruszyńska, Katarzyna et al. (*incl. Abrams, N. S.*). “Synergies between Roman Galactic Plane Survey and other major surveys”. *arXiv e-prints*. arXiv:2406.14767. 2024

11. Terry, S. K., et al. (*incl. Abrams, N. S.*) “The Galactic Center with Roman” *Roman Core Community Survey White Paper*. June 2023.
10. Lam, Casey Y. et al. (*incl. Abrams, N. S.*). “Roman CCS White Paper: Characterizing the Galactic population of isolated black holes”. *Roman Core Community Survey White Paper*. arXiv:2306.12514. 2023
9. Street, R. A. et al. (*incl. Abrams, N. S.*). “Maximizing science return by coordinating the survey strategies of Roman with Rubin, and other major facilities”. *Roman Core Community Survey White Paper*. arXiv:2306.13792 *Roman Core Community Survey White Paper*. 2023
8. ***Abrams, N. S.**, et al., “Microlensing Discovery and Characterization Efficiency at Different Timescales” *Rubin Observatory Cadence Note*. April 2021.
7. Hundertmark, M., et al. (*incl. Abrams, N. S.*) “Alerting transient phenomena in the Galactic Plane in time to coordinate follow-up” *Rubin Observatory Cadence Note*. April 2021.
6. Street, R., et al. (*incl. Abrams, N. S.*) “LSST Survey Footprint in the Galactic Plane and Magellanic Clouds” *Rubin Observatory Cadence Note*. April 2021.
5. Bachelet, E., et al. (*incl. Abrams, N. S.*) “On the observational synergies between all-sky surveys for the characterization of microlensing events” *Rubin Observatory Cadence Note*. April 2021.
4. Dawson, W., Smyth N., et al. (*incl. Abrams, N. S.*) “Rubin/LSST Black Hole Dark Matter Microlensing” *Snowmass2021 - Letter of Interest*. August 2020.
3. * **Abrams, N. S.**, Gomez, S., and Bieryla, A. “Measured Lightcurves and Rotational Periods of (16579) 1992 GO (25660) 2000 AO88, And (37652) 1994 JS1.” *Minor Planet Bulletin*. July 2020. 168-169.
2. ***Abrams, N. S.**, et al., “Measured Lightcurves and Rotational Periods of 3122 Florence, 3830 Trelleborg, and (131077) 2000 YH105.” *Minor Planet Bulletin*. January 2020. 3-4.
1. ***Abrams, N. S.**, “Galaxy Morphology Dependence on Mass and Luminosity.” Proceedings of 24th International Competition “First Step to Nobel Prize in Physics”(FSNPP) 2016. Fall, 2016. 294-306.

PRESENTATIONS (9 invited talks, 29 contributed talks, 3 posters)

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| 41. LSST-Europe8 | Sept 2026 |
| - Invited talk: “Roman and Rubin Microlensing Synergize to Characterize Black Holes and Binary Stars” | |
| 40. Rubin Community Workshop 2026 | July 2026 |
| - Contributed talk: “Discovering and Interpreting Microlensing events with Rubin via ANTARES” | |
| 39. Earth 2.0 Science Meeting | July 2026 |
| - Invited talk: “Discovering Microlensing with Rubin LSST” | |
| 38. Roman Shifting Landscapes in Astrophysics | July 2026 |
| - Contributed talk: “Illuminating Hidden Binary Parameter Space with Roman Microlensing” | |
| 37. MIT Burdge Group Meeting | June 2026 |
| - Contributed talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys” | |
| 36. UC Berkeley Astronomy Lunch talks | April 2026 |
| - Contributed talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys” | |
| 35. Carnegie Observatory Lunch Seminar | April 2026 |
| - Invited talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys” | |
| 34. Caltech Tea talk | April 2026 |
| - Invited talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys” | |
| 33. CCA Nearby Universe Group | April 2026 |
| - Contributed talk: “Probing Binaries and Black Holes with Photometric and Astrometric Microlensing” | |
| 32. CCA Coffee talks | April 2026 |
| - Invited talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys” | |

31. Columbia Time Domain Meeting March 2026
- Contributed talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys”
30. Carnegie Mellon Binary Group Meeting March 2026
- Contributed talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys”
29. International Microlensing 28 Conference Feb 2026
- Invited talk: “Rubin Microlensing and the Search for Black Holes and Binaries”
28. KITP The Lifecycle of Stellar Black Holes Nov 2025
- Poster: “Gravitational Microlensing in the Era of All-Sky Surveys”
27. Las Cumbres Observatory Science Seminar Nov 2025
- Invited talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys”
26. UC Santa Barbara Astro Lunch Nov 2025
- Contributed talk: “Probing Binaries and Black Holes with Microlensing in the Era of All-Sky Surveys”
25. Roman RGS PIT Meeting Oct 2025
- Contributed talk: “Roman Filters, Binary Orbital Motion, and other updates in PopSyCLE”
24. Keck Science Meeting Sept 2025
- Poster: “Upgrading Keck AO Imaging Pipeline (KAI) to Analyze Candidate Astrometric Microlensing Black Holes”
23. Rubin Community Workshop August 2025
- Contributed talk: “Searching for Black Holes with Rubin Microlensing”
22. Northwestern Transients Meeting June 2025
- Contributed talk: “Gravitational Microlensing in the Era of All-Sky Surveys”
21. UC Berkeley Astronomy Lunch talks Feb 2025
- Contributed talk: “Assessing the Impact of Binary Systems on Microlensing Using SPISEA and PopSyCLE Population Simulations”
20. International Microlensing 27 Conference Jan 2025
- Contributed talk: “Bias in Fitting Simulated Microlensing Lightcurves from PopSyCLE”
19. Ciela Institute’s Astromerique Student talk Series Nov 2024
- Invited talk: “Gravitational Microlensing in the Era of All-Sky Surveys”
18. Roman PIT October 2024
- Contributed talk: “Using PopSyCLE to Assess Binaries in Roman Microlensing”
17. Rubin Community Workshop July 2024
- Contributed talk: “Microlensing in the Era of All Sky Surveys”
16. Challenging Theory with Roman: From Planet Formation to Cosmology July 2024
- Contributed talk: “Assessing the Impact of Binaries on Microlensing in PopSyCLE”
15. International Microlensing 26 Conference Jan 2024
- Contributed talk: “Microlensing in the Era of All Sky Surveys”
14. 243rd American Astronomical Society Jan 2024
- Contributed talk: “Microlensing in the Era of All-Sky Surveys”
13. Rubin Project and Community Workshop Aug 2023
- Contributed talk: “Microlensing Discovery and Characterization Efficiency in the Vera C. Rubin Legacy Survey of Space and Time”
12. UC Berkeley Astronomy Short talks March 2023
- Contributed talk: “60 Microlensing Events in ZTF One”
11. International Microlensing 25 Conference Aug 2022
- Contributed talk: “Assessing the Impact of Binary Systems on Microlensing”

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| 10. UC Berkeley Astronomy Short talks | March 2022 |
| - “Assessing the Impact of Binary Systems on Microlensing” | |
| 9. Rubin Observatory Project and Community Workshop | Aug 2021 |
| - Contributed talk: “Microlensing Discovery, Alerts, and Characterization Efficiency at Different Timescales in the Vera C. Rubin Legacy Survey of Space and Time” | |
| 8. 238th American Astronomical Society | June 2021 |
| - Contributed talk: “What’s Hiding Amongst the Pulses?: Using Phase Modulation in the Light Curves of RR Lyrae Variables to Search for Black Holes” | |
| 7. 237th American Astronomical Society | Jan 2021 |
| - Contributed talk: “Assessing the Effect of Binary Systems on Microlensing Adding Binary Systems to SPISEA and PopSyCLE” | |
| 6. UC Berkeley Astronomy Short talks | Oct 2020 |
| - Contributed talk: “Hunting black holes with photometric microlensing” | |
| 5. Rubin Observatory Project and Community Workshop | Aug 2020 |
| - Contributed talk: “Hunting gravitational wave black holes with microlensing” | |
| 4. San Francisco State University Dark Matter Series Guest Lecture | Aug 2020 |
| - Invited talk: “MACHOs” | |
| 3. 235th American Astronomical Society | Jan 2020 |
| - Contributed talk: “Assessing LSST’s ability to hunt LIGO black holes with microlensing” | |
| 2. Harvard Summer Undergraduate Research Village | Aug 2018 |
| - Contributed talk: Probing the evolution of supermassive black holes in various galaxy environments | |
| 1. 232nd American Astronomical Society | June 2018 |
| - Poster: “Developing methods of determining unknown rotational periods of asteroids via observations of (3122) Florence by the Harvard Observing Project” | |

HONORS/AWARDS

Berkeley Fellowship	2021 – 2026
Berkeley Astronomy DEIC Award	2025
Hearts to Humanities Eternal	2025
UC Berkeley Outstanding Graduate Student Instructor	2023
NSF Graduate Research Fellowship Program Honorable Mention	2022
LSSTC Enabling Science Award	2021 – 2022
John Harvard Scholar	2019 – 2020
Herchel Smith Fellowship	Summer 2020
Harvard Undergraduate Science Research Program	
2020 AstroTech Summer School	Summer 2020
Japan Summer Science Undergraduate Research Program	Summer 2019
Program for Research in Science and Engineering (PRISE)	Summer 2018

TEACHING EXPERIENCE

Instructor: Akamai PREP Course (Camera Obscura)	Summer 2024
Professional Development Program 2024	
Graduate Student Instructor: Astronomy Data Science Laboratory (Astronomy 128/256)	Fall 2022
University of California, Berkeley (Prof. Jessica Lu & Prof. Aaron Parsons)	
Graduate Student Instructor: Introduction to Astrophysics (Astronomy 7B)	Spring 2022
University of California, Berkeley (Dr. Ryan Chornock)	
Course Assistant: Nonlinear Dynamical Systems (Applied Math 108)	Fall 2020
Harvard University (Dr. Sarah Iams)	

RESEARCH MENTORSHIP

Primary graduate student advisor for UC Berkeley undergraduates advised by Prof. Jessica Lu:

- T. Dex Bhadra - Adding orbital motion to BAGLE Microlensing code (Published 2026) Fall 2023 - Present
- Madelyn Rahimi - Simulating ZTF microlensing events for PopClass Summer 2025 - Present
- Sage Hironaka Remulla - Adding Rubin and Roman filters to PopSyCLE Fall 2024 - Present
- Kai Marshall - Improving PopSyCLE event calculation algorithm Summer 2024
- Abby Schleigh - Improvements to BAGLE documentation and code Fall 2023 - Spring 2024

OBSERVING EXPERIENCE

Keck Observatory

- NIRSPEC w/ LGS AO Co-I (Major Contribution): 1 night (Cycle 25A)
- OSIRIS Imager w/ LGS AO Co-I (Major Contribution): 12.5 nights (Cycles 22A-26B)

Lick Observatory

- Shane Kast **PI:** 10 nights (Cycles 24B-26A)
- Automatic Planet Finder Co-I (Minor Contribution): 180 hr (Cycles 23A-23B)
- Graduate Popper Workshop 2023

JWST

- NIRCam Imaging Co-I (Major Contribution): 4.1 hours (Director's Discretionary 2024-2025)
Co-I (Major Contribution): 10.8 (Cycle 5-7)

Fred Lawrence Whipple Observatory

- 60" telescope (KeplerCam) 5 nights, 2019
- 48" telescope (FAST) 2 nights, 2019

SERVICE

Outreach

- UC Berkeley Astro Night Organizer 2022 – Present
- UC Berkeley Outreach Coordinator 2022 – Present
- Wonderfest Science Envoy 2024 – 2025
- UC Berkeley Society for Women in Physical Sciences (SWPS) Mentor 2023 – 2025
- UC Berkeley Compass Mentor 2021 – 2022
- Harvard Society for Physics Students (SPS) Mentor 2019 – 2021
- Harvard Observing Project (HOP) Instructor 2018 – 2021
- Orchestar: Color Sonification Arduino Developer 2017
Device to make observational astronomy accessible to people with visual impairments.
- 16 public talks in the Bay Area and Santa Barbara

Institutional Service

UC Berkeley Astronomy Department

- Graduate Representative for Faculty Search 2026
- Graduate Representative for Graduate Admissions 2026
- Graduate Meeting w/ Colloquium Speaker Organizer and Facilitator 2022 – Present
- Graduate Peer Mentor 2023 – 2025

Harvard Astronomy Department

- Harvard Undergraduate Astronomy Society (AstroSoc) 2019 – 2021
Co-President and Founder

Conference and Workshop Organizing Committees

- LOC, Microlensing 26 2024

SKILLS

Programming Languages: Python; Bash

Software and Frameworks: Cursor; LaTeX; Git; Unix; Slurm; SAOImage DS9; IRAF Command Language

Languages: English (Native), Spanish (Conversational), Japanese (Conversational)